Formation, Education, and Training in SymK

How durable change, Learning, Training and Education remain distinct

SymK

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Contents

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Purpose and audience

Purpose. This paper explains why Formation is the foundational concept while Education, Training and Learning remain distinct. It is designed to be useful to newcomers, educators, system designers and governance readers without requiring the evolution history as a prerequisite.

Audience. People concerned with Human development, model or organizational adaptation, training programs, cooperative practice and the evaluation of change. No specialist educational theory or SymK background is assumed for the entry path.

The question this paper answers. How should SymK distinguish durable change, Learning, Training and Education without assuming that every change is beneficial or that Cooperation follows automatically?

Authority and limits. This is a governed explanatory paper, not a constitutional norm, final semantic package, formal specification or implementation contract. Exact Stage-Accepted decisions control if wording here and those decisions differ.

SymK in plain language

SymK is a general framework and engineering discipline for responsible Knowledge Engineering. It helps people and other capable systems work together on difficult questions while keeping claims, evidence, authority, responsibility, use, consequences and correction visible.

In ordinary language, Knowledge Engineering is the disciplined work of making Domain knowledge expressible, organized, connected to evidence, reviewable, maintainable, challengeable and usable by people and computational systems. It may use vocabularies, ontologies, knowledge bases, models, workflows and AI, but no artifact or technology is the discipline by itself.

Within SymK, Knowledge Engineering is the integrative center and epistemic conditions are its direct engineering object. This paper supplies Formation as the constitutive account of comparatively durable change across people, systems and organizations. Education and Training are conditional formative practices: they matter when capabilities or participants are deliberately shaped, but neither is a universal requirement of every SymK undertaking. Future supporting concepts may be admitted through governed evidence without collapsing these distinctions.

SymK began with Human-AI cooperation and still treats Domain Intelligence Amplifiers as a principal realization. Its scope is now broader: it can examine governed cooperation among different forms and Bearers of Intelligence. This broader identity does not make every activity part of SymK, erase Human standing or give a technical system authority merely because it is capable.

Why this paper matters. A person, model, team or institution can change without improving. A curriculum can be delivered without Education occurring. A model can be updated without the larger system learning. SymK requires the subject, process, effect, evidence, value and authority claims to be separated so that beneficial development is not merely assumed.

Two reading paths

Short path - about fifteen minutes. Read this opening, the Abstract, section 1 and the Conclusion. That path states the problem, the central answer, the practical consequence and the limits without requiring the full evolution history.

Full path - for deeper knowledge. Continue through every numbered section, including the counterexamples, distinctions, engineering implications, dissent and references. Those sections support scrutiny and specialist use; they are not prerequisites for understanding the main claim.

The optional *SymK Reader Orientation and Shared Glossary* introduces recurring terms across all four Foundation Papers. This paper nevertheless defines the terms required for its own main argument.

One example to carry through the paper

After the bridge warning, the organization changes its inspection routine, engineers practise a new method and the model is updated. These are formative conditions and activities. They do not by themselves prove Learning, successful Training or Education. Evidence must identify what changed, in which Learner or organized system, how durable and transferable the change is, and whether it was beneficial, harmful or mixed. Cooperation may improve, but it is not guaranteed by the intervention.

The example is illustrative. It does not create a universal workflow, legal conclusion or operational rule.

Essential terms for this paper

- **Formation:** comparatively durable shaping or reorganization of an identified subject through experience, intervention, participation, environment or self-directed activity.
- **Learning:** comparatively durable, experience-linked change at an identified Learner level.
- **Training:** directed and structured formative activity toward declared performance or disposition conditions.
- **Education:** broad, developmental, integrated and challengeable Formation connecting Knowledge, judgment, conduct, relationship and revision of the frame.
- **Cooperation:** a thin relational process; it is not automatically good, legitimate, successful or the final objective.
- **Evidence:** support for a claim about process, effect or achievement; participation or completion alone is not proof.

Capitalization marks terms that have a controlled SymK use. These short explanations support reading; the Stage-Accepted decisions remain the controlling conceptual source.

Abstract

SymK is a general framework and engineering discipline for responsible Knowledge Engineering through governed cooperation among different forms and Bearers of Intelligence. Cooperation cannot be assumed simply because participants are knowledgeable or capable. People, models, teams and institutions may need to develop, adapt or reorganize - but not every change is Learning, not every directed activity is successful Training, and not every broad developmental process is Education.

The foundational concept is **Formation**: the temporally extended shaping or reorganization of an identified subject's comparatively durable capacities, dispositions, practices, judgments, relationships or organized patterns through experience, intervention, participation, environment or self-directed activity.

Formation is value-neutral. It may be intended or unintended, beneficial or harmful, integrative or deformative. A curriculum, interface, repeated interaction, model update, organizational routine or stored trace may create formative conditions; none proves that Formation occurred or that its result was good. SymK therefore separates purpose, condition, exposure, activity, mechanism, effect, achievement, attribution, assessment, status and record.

Learning, Training and Education are distinct supporting concepts. They may overlap, but none universally entails the others. The proposed Learner level must be identified: component change is not automatically system Learning, and collective change requires evidence at the collective level. Cooperation remains a thin relational process rather than a guaranteed result or moral endorsement.

This paper develops those distinctions, preserves concerns about habit, power, hidden curricula, assimilation, reciprocal influence, decay, challenge and repair, and explains how evidence and correction should govern claims about formative activity and effect.

Review status and source control

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1. Why Formation is foundational

SymK's layered identity preserves its historical Human–AI symbiotic-cooperation protocol, its principal but non-exhaustive orientation toward Domain Intelligence Amplifiers, and its broader identity as an engineering discipline and general framework for responsible Knowledge Engineering through governed Cooperation among heterogeneous forms and Bearers of Intelligence. These layers are related but not identical.

The Knowledge foundation distinguishes Knowledge from claims, Representations, records, status, and storage. The Intelligence foundation treats Intelligence as an actual, substrate-realized capacity of an organized system rather than a material component, isolated performance, score, or human resemblance. The present paper asks a different question:

> How can an identified subject be shaped over time, and when may the result be called Learning, Training achievement, Education, improvement, or cooperative fitness?

The general answer is Formation. A subject may acquire or lose capacities, dispositions, practices, judgments, relationships, and organized patterns. That shaping can occur through declared instruction, structured practice, environmental exposure, participation, self-directed activity, institutional routines, coercion, redesign, or combinations of these. It can support responsible participation, or it can produce dependency, prejudice, concealment, brittleness, domination, and deskilling.

This neutral foundation matters because evaluative vocabulary can hide what happened. Calling an intervention "education" may confer undeserved legitimacy. Calling a model-development process "training" does not prove any achieved capability. Calling a changed workflow "organizational learning" may conceal replacement, selection, retrieval, supervision, or environmental redesign. Formation permits the actual shaping to be investigated before its mechanism, achievement, value, or status is decided.

Intelligence still does not entail cooperative fitness. Knowledge still does not entail responsible conduct. But neither fact makes Education a universal prerequisite of Cooperation. Cooperation is an actual, scoped relational process in which two or more distinguishable Participants make materially relevant Contributions that are mutually oriented toward a bounded shared undertaking and sufficiently connected for the Contributions to form part of that undertaking rather than merely coincide or aggregate. It may occur without Education, and Education may occur without Cooperation in a particular episode. Stable capability continuity may be supported through many paths, including selection, replacement, redesign, safeguards, supervision, institutional memory, practice, Training, Learning, Education, or environmental change.

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2. The distinctions SymK must preserve

2.1 Formation

> **Formation is the temporally extended shaping or reorganization of an identified subject's comparatively durable capacities, dispositions, practices, judgments, relationships, or organized patterns through experience, intervention, participation, environment, or self-directed activity.**

Formation is broader than Learning, Training, or Education. Its subject may be human, artificial, collective, institutional, distributed, or hybrid if that level has a stable enough boundary and adequate Evidence. The term does not decide whether the process was intended, effective, beneficial, legitimate, consensual, or educational.

2.2 Learning

> **Learning is comparatively durable change at an identified Learner level linked to experience, inquiry, practice, feedback, interaction, or adaptation.**

Learning is an effect claim, not the name of every activity intended to produce it. A person may learn avoidance; a team may learn to conceal failure; a configured system may learn a more reliable procedure. Harmful patterns can be learned. A new output, retrieved record, selected member, replaced component, or externally imposed instruction does not by itself show Learning.

2.3 Instruction

Instruction is guided presentation, explanation, demonstration, questioning, task setting, or support intended to create conditions for Learning or another formative effect. Instruction is something an instructor, peer, environment, or system does. Reception, uptake, Learning, achievement, and current capability require separate Evidence.

2.4 Training

> **Training is directed and structured formative activity toward declared performance or disposition conditions.**

Training commonly uses practice, feedback, correction, rehearsal, reinforcement, variation, and assessment. Repetition may be one method, but it is neither universally necessary nor sufficient. A Training activity may fail. A trained achievement, transfer to unfamiliar conditions, durability after support is removed, and current status are separate claims.

2.5 Education

> **Education is broad, developmental, integrated, participatory, and challengeable Formation connecting Knowledge, judgment, conduct, relationship, and capacity to examine and revise the frame.**

Education is more than content delivery, isolated skill acquisition, credentialing, or compliance. It concerns the organization of participation, Knowledge, judgment, conduct, relationship, and revision over time. It remains an attributed formative achievement at an identified subject level, not a property conferred by the name of a school, curriculum, institution, interface, or program.

2.6 Socialization, adaptation, conditioning, and practice

Socialization is Formation through participation in social practices and expectations. It may enable membership or reproduce domination and exclusion. Education therefore cannot be equated with successful integration into the existing order.

Adaptation is change relative to conditions. It may be transient, externally selected, harmful, or not linked to experience at the claimed Learner level.

Conditioning is shaping through patterned contingencies. It may produce stable conduct without the integrated and challengeable organization required for Education.

Practice is repeated or continuing activity through which a capacity or disposition may be exercised. Practice is broader than Training: it can be self-directed, informal, incidental, unstructured, culturally embedded, or exploratory.

2.7 Comparative view

Accessible table projection: Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **Concept:** Formation
  - **Predicate at issue:** comparatively durable shaping or reorganization
  - **What does not establish it automatically:** exposure, architecture, purpose, activity, repetition, record
  - **Separate questions:** subject, mechanism, effect, direction, value, attribution
- **Concept:** Learning
  - **Predicate at issue:** comparatively durable experience-linked change at a Learner level
  - **What does not establish it automatically:** output difference, retrieval, replacement, instruction, feedback
  - **Separate questions:** durability, linkage, manifestation, alternatives, current status
- **Concept:** Instruction
  - **Predicate at issue:** guided formative activity
  - **What does not establish it automatically:** content delivery or instructor intent alone
  - **Separate questions:** access, reception, uptake, effect, authority
- **Concept:** Training
  - **Predicate at issue:** directed structured formative activity toward declared conditions
  - **What does not establish it automatically:** repetition, industrial terminology, completed course
  - **Separate questions:** objective, structure, participation, success, transfer, decay
- **Concept:** Education
  - **Predicate at issue:** broad integrated participatory challengeable Formation
  - **What does not establish it automatically:** schooling, credential, compliance, Training, architecture
  - **Separate questions:** Knowledge, judgment, conduct, relationship, frame revision, power
- **Concept:** Socialization
  - **Predicate at issue:** Formation through social practices and expectations
  - **What does not establish it automatically:** membership or conformity alone
  - **Separate questions:** what order, whose norms, burdens, alternatives, challenge

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3. Formation activity, effect, achievement, and attribution

SymK must not collapse a formative sequence into one label. At least the following distinctions matter:

- a **formative purpose** states an intended developmental direction;
- a **formative condition** can influence a subject;
- a **formative environment** organizes multiple conditions;
- **formative exposure** places the subject in contact with them;
- a **formative activity** is directed or functions toward shaping;
- a **formative mechanism** explains how conditions connect to effects;
- a **formative effect** is actual shaping at the identified subject level;

- a **formative achievement** is an effect satisfying declared criteria;
- a **formative attribution** is a fallible claim about condition, mechanism, effect, or achievement;
- an **assessment** evaluates such a claim;
- a **status** records a governed current classification; and
- a **record** represents the purpose, activity, Evidence, assessment, status, or lineage.

Each transition can fail. An intervention may be offered but not received. Exposure may occur without uptake. Uptake may be temporary. A durable change may not transfer. An achievement may decay. A record may outlive the condition it describes. A competent authority may recognize a status without creating the underlying effect, and an actual effect may exist before any authority recognizes it.

Value also remains separate. Formation may be beneficial or harmful, integrative or deformative, emancipatory or assimilative, intended or unintended. A declared educational purpose does not confer legitimacy. A beneficial effect does not retrospectively authorize coercive means. A stable disposition is not automatically a virtue.

4. Education and Training: contingent relations, not a universal axiom

The v0.1 paper argued that Education needs Training and proposed an Axiom of Cooperative Formation. That inquiry correctly identified a practical risk: principles that never become available in conduct can be fragile, while repeated habits without understanding can become conformity. It incorrectly promoted one possible relation into a universal dependency.

The accepted baseline is narrower and stronger:

- Education and Training may overlap in one episode or developmental history;
- Training may contribute to Education when structured practice supports integrated, participatory, and challengeable Formation;
- Education may guide the purpose, interpretation, limits, and revision of Training;
- Training can occur without Education;
- Education can occur without separately directed repeated Training;
- practice can support Education without qualifying as Training;
- Learning may result from Training, Education, both, or neither;
- Formation may occur without Learning, Training, or Education being established; and
- none is universally necessary or sufficient for Cooperation, persistence, system Learning, improvement, or responsible development.

Dewey’s treatment of habit remains useful historical and philosophical evidence that acquired dispositions organize conduct and inquiry [Dewey 1922; Fesmire 2024]. It does not decide SymK semantics or justify a universal Education–Training co-entailment. Deliberate practice, varied cases, reflection, exemplars, feedback, environmental design, social participation, and institutional change are plural possible mechanisms. Their actual effects remain empirical and scoped.

The proposed Axiom of Cooperative Formation is therefore preserved only as historical reasoning from v0.1. This paper does not advance it as a candidate axiom or constitutional proposition. A narrower duty concerning responsible or evolvable system capability would require separate constitutional ownership and procedure.

5. Education, participation, and a shared world

Education is frequently reduced to schooling, curriculum, content transmission, credentialing, or occupational preparation. These are institutions, activities, or projected purposes; they are not its universal identity.

Dewey linked Education with experience, inquiry, democracy, and associated life [Dewey 1916]. The Delors report described learning to live together among four educational pillars [Delors et al. 1996]. These sources illuminate traditions in which Education concerns participation in a world inhabited and interpreted by others. They remain literature, not semantic or constitutional authority for SymK.

5.1 Integration is not assimilation

Participation often requires shared forms: language, turn-taking, symbols, procedures, expectations, and boundaries. Yet integration must not be inferred from compliance, and it must not erase relevant difference. Existing orders can socialize domination, exclusion, silence, or dependence. Education remains challengeable precisely because the frame, educator, curriculum, classification, and institution may be wrong.

5.2 Relational practices are contextual

Listening, acknowledging receipt, identifying uncertainty, respecting privacy, distinguishing criticism from humiliation, restraint, and repair can reduce friction in many cooperative settings. But no universal virtue catalogue, etiquette, culture, or actor topology follows. A proposed habit must be assessed by Domain, Scope, Context, role, consequence, burden, exclusion, and authority.

5.3 Environments can be formative without being educators

Interfaces, incentives, defaults, permissions, memory systems, examples, sanctions, and tolerated violations can create formative conditions. Designers should investigate the patterns these conditions encourage. But an architecture does not literally educate or train merely because it can influence conduct. Actual effect, Learner level, mechanism, durability, transfer, alternative explanations, and appropriate predicate must be evidenced.

The phrase “hidden curriculum” remains useful for investigating undeclared formative conditions. It is not a license to attribute Education wherever environmental influence is suspected.

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6. When Learning, Training achievement, and Education may be attributed

6.1 Learning

A Learning claim should identify:

- a stable enough Learner boundary;
- relevant prior and later conditions;
- experience, inquiry, practice, feedback, interaction, or adaptation linked to the change;
- comparatively durable integrated change at that Learner level;
- manifestation under relevant variation; and
- control of plausible alternatives such as selection, replacement, redesign, current Context, retrieval, supervision, or environmental change.

Learning need not be beneficial or reflective. Its status must remain current and revisable in light of decay, drift, unlearning, and new Evidence.

6.2 Training and trained achievement

A Training activity is established by a declared target and directed structure, not by eventual success. A trained-achievement claim additionally needs Evidence of the relevant performance or disposition under stated conditions, with durability, dependence, variation, transfer, failure, and decay made visible.

Machine “training,” fine-tuning, reinforcement, prompting, retrieval augmentation, and adaptation must be described at the actual level and mechanism. Parameter change may affect outputs without establishing Education, transfer, stable capability, model-level Learning, or system Learning. Industrial terminology does not decide the predicate.

6.3 Education

An Education attribution requires a wider developmental profile connecting:

- Knowledge and the ability to distinguish claims, Grounds, Evidence, Representation, and status;
- judgment where rules underdetermine action;
- conduct under relevant variation;
- participation and relationship;
- capacity to receive, evaluate, and incorporate correction;
- recognition of conflict among purposes, norms, interests, and consequences; and
- practical capacity to examine and revise the frame through governed challenge.

No single performance proves Education. These considerations are a review envelope, not a universal score, certification workflow, or schema. Domain and external authorities retain curricula, professional standards, credentials, legal obligations, pedagogies, and thresholds within jurisdiction.

6.4 One episode, several predicates

The same episode may include Instruction, practice, Training, Learning, and Education, but each predicate requires its own subject, time, mechanism, effect, and Evidence. A repeated exercise in fair restatement can be Training. If an identified subject later exhibits comparatively durable change linked to that practice, Learning may be attributed. If the change becomes part of broad, integrated, participatory, and challengeable Formation, it may contribute to Education. No label follows automatically from another.

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7. Human, artificial, collective, institutional, distributed, and hybrid cases

Formation is substrate-neutral but not substrate-indifferent. The relevant mechanism, boundary, duration, manifestation, dependence, and alternatives differ across cases.

7.1 Human subjects

Human Formation is shaped by embodiment, development, emotion, language, culture, dependency, mortality, trust, belonging, institutions, and material conditions. Human origin does not imply completed Education, cooperative fitness, or freedom from harmful Formation. Human dignity and external moral or legal standing do not depend on Intelligence, Knowledge, educational achievement, or cooperative performance.

7.2 Artificial systems

Artificial systems can undergo parameter modification, rule change, memory change, retrieval reconfiguration, feedback-driven adaptation, tool substitution, or workflow redesign. Each has a possible Learner and mechanism boundary. Artificial Education remains a literal claim only when the integrated, participatory, challengeable predicate is supported at the identified level; otherwise “education” is metaphorical, aspirational, or misapplied.

7.3 Collective and institutional subjects

A team, institution, or community may change its roles, routines, information flow, challenge mechanisms, memory, incentives, and repair practices. That does not automatically make the whole a Learner or Education Bearer. Member Learning does not transfer upward, and system Learning does not transfer downward.

System Learning requires a stable Learner boundary, system-level experience and mechanism, integrated comparatively durable change, manifestation under relevant variation, and control of selection, replacement, redesign, Context, retrieval, and environmental explanations. Use the lowest adequate Bearer: do not claim a system-level predicate when a component-, member-, or workflow-level explanation is sufficient.

7.4 Distributed and hybrid subjects

A Human–AI or other distributed arrangement may develop a capability through coupling even while some components remain fixed. The actual change may lie in role allocation, interfaces, checking practices, memory, authorization, escalation, or feedback. Evidence must identify whether the Learner is a person, model, tool, team, organization, configured system, or another bounded subject. Replacement and succession conditions must be explicit.

8. Reciprocal Formation, persistence, and memory

Participants in continuing relations can shape one another. Reciprocal Formation need not be symmetric, beneficial, educational, or visible at every level. One participant may change while another does not. An environment may be redesigned while members retain prior dispositions. A fixed component may support a wider system's Learning without learning itself.

Temporal objects must remain distinct:

- a cooperative episode;
- repeated interaction;
- a recurring pattern;
- a continuing cooperative relation; and
- a persistent system under declared Identity conditions.

Duration, repetition, stable name, unchanged membership, storage, or organizational status does not establish persistence. Persistence does not establish Education or Training.

Stored trace, retrieval, current Context, procedural retention, institutional memory, distributed memory, developmental history, and actual Learning also differ. A record can support later Learning without being memory in every sense; retrieval can change current performance without durable change; institutional continuity can preserve a harmful practice; and capability may decay while a status record remains unchanged.

9. Power, coercion, affected subjects, and challenge

Formation is never explained adequately by content and mechanism alone when power is material. Whoever selects objectives, examples, rewards, penalties, access, timing, evaluation criteria, and acceptable forms of dissent can shape participants and distribute burdens.

Power capacity, exercise, mechanism, effect, qualification, Authority, governance response, and record remain distinct. Training may be compulsory without that fact alone deciding legitimacy; Education can become indoctrination or assimilation; informal socialization can be more coercive than declared instruction. Consent, necessity, dependence, vulnerability, alternatives, reversibility, transparency, and consequence require Domain and external authority.

Formal permission to question a frame is not effective challengeability. Material review should distinguish notice, access, practical capacity, protection, standing or representation, reception, evaluation, reasoned response, possible corrective consequence, repair or remedy interface, appeal, recurrence, and Learning.

Affected-subject standing/challenge is an unresolved locus distinct from the eleven typed Authority kinds. It is not automatically an Authority kind, veto, legal standing, participation right, duty, remedy, or decision power. SymK must preserve the locus and the affected horizon without claiming jurisdiction to allocate external moral, legal, professional, democratic, community, or institutional standing.

Evidence may challenge claims, premises, Applicability, consequences, and decisions operating under any Authority. It does not automatically amend a governed artifact, alter or revoke an Authority or its source, or reverse a decision; such change follows competent procedure. Urgent adverse Evidence triggers urgent consideration by competent operational or external Authority. This creates no emergency duty, containment or suspension power, decision right, remedy, or jurisdiction.

10. Cooperation and responsible qualification

Cooperation is a thin relational occurrence, not moral endorsement. It can be helpful, harmful, coerced, exploitative, collusive, unequal, unstable, or irresponsible. Education and Training may influence a participant's contributions, but neither guarantees Cooperation, responsibility, legitimacy, Value, or success.

Participant, Contributor, actor, resource, source, beneficiary, affected party, Bearer, authority, and responsible party are distinct roles. A cooperative unit under analysis is not automatically a Participant, Intelligence Bearer, Learner, legal person, authority, or responsible party.

The v0.1 habits—recognition, attentive reception, epistemic transparency, interpretive charity with resistance, restraint, correctability, repair, reciprocity without symmetry, and reflective revision—remain useful paper-level proposals for contextual assessment. They are not universal virtues, constitutional duties, participant-admission criteria, or canonical schema. A Domain may justify, revise, add, or reject them within competent jurisdiction while preserving SymK's semantic minima.

Cooperation is not the final objective of SymK. Questions can lead through Reasoning to provisional Answers; Answers can support decisions and actions; use can produce multiple outcomes; outcomes can have different Value for different subjects. Observations and claims about consequences can be assessed for Evidence roles, motivate Correction, change the Question, or support ending an undertaking; none does so automatically. Formation participates in this branching and recursive system without guaranteeing a preferred branch.

11. Explanatory obligations replacing candidate commitments

The v0.1 `E-01`–`E-12` catalogue is retained as historical Proposed reasoning, not as candidate constitutional commitments. Its supported content is reconstructed as the following explanatory obligations:

1. **Non-entailment:** Intelligence, Knowledge, capability, origin, status, or Authority does not establish cooperative fitness, Learning, Training, Education, or responsible conduct.
2. **Distinct concepts:** Formation, Learning, Instruction, Training, Education, Socialization, Adaptation, Conditioning, and practice must not be silently substituted.
3. **No universal co-entailment:** Education and Training can overlap and support one another, but neither universally requires or entails the other.
4. **Activity/effect separation:** formative purpose, conditions, activity, mechanism, effect, achievement, attribution, status, and record require separate treatment.
5. **Value neutrality of Formation:** Formation and Learning can be harmful; repetition can entrench error, dependency, prejudice, coercion, or brittle optimization.
6. **No architecture effect:** an interface, institution, curriculum, persistent arrangement, or model update can create formative conditions but does not automatically train, educate, or produce Learning.
7. **Challengeability:** Education connects participation with practical capacity to examine and revise the frame; formal permission alone is insufficient.
8. **Power and affected horizon:** objectives, methods, Evidence, burdens, alternatives, coercion, affected subjects, and consequences must remain inspectable.
9. **Scoped attribution:** every Learning, trained-achievement, Education, or system-level claim must identify subject, level, time, mechanism, Evidence, alternatives, durability, transfer, and current status proportionately to consequence.
10. **No cross-level transfer:** component, member, system, institutional, distributed, and hybrid predicates require their own boundaries and Evidence.
11. **Temporal honesty:** persistence, memory, continuity, decay, drift, deskilling, recovery, and succession must not be hidden by a permanent label.

12. **Jurisdiction:** SymK owns universal semantic minima; Domain and external authorities own actual curricula, standards, methods, credentials, coercion rules, thresholds, duties, rights, remedies, and decisions.

These obligations explain accepted decisions. They do not constitute an axiom set or allocate Authority.

12. Counterexamples that the account must survive

1. A knowledgeable participant knowingly acts deceptively. Knowledge does not entail responsible conduct or Education.
2. Two uneducated participants make mutually oriented Contributions to a bounded undertaking. Cooperation can occur without Education.
3. A person develops broad, reflective judgment through participation and inquiry without a separately directed repeated Training program. Education can occur without Training.
4. A system completes Training but shows no durable target improvement. Training activity does not entail trained achievement.
5. A repeated exercise entrenches an error. Repetition does not entail Learning of the intended pattern or beneficial Formation.
6. A learner develops a harmful avoidance response. Learning and Formation are value-neutral.
7. An interface rewards speed and concealment, but no durable user-level change is shown. A formative condition is not an established effect.
8. A model update changes parameters and benchmark output but not transfer under relevant variation. Parameter change does not prove stable capability or Education.
9. A team improves because one specialist is replaced. Better system performance does not prove system Learning.
10. Members complete a course while institutional incentives remain unchanged. Member activity does not establish institutional Education.
11. A workflow changes while its fixed model does not. The configured system may change without model-level Learning.
12. A stored procedure is retrieved successfully. Retrieval does not establish current Learning or institutional memory in every sense.
13. A stable cooperative relation becomes exploitative. Persistence and Cooperation do not establish legitimacy or Value.
14. Formal challenge is available, but dissent causes retaliation. Permission does not establish effective challengeability.
15. Evidence shows severe harm from an educational program. Evidence challenges the program and decisions; it does not itself amend the governing authority source.
16. An affected nonparticipant bears costs from a successful Training program. Participant achievement does not settle standing, Responsibility, remedy, or Value.
17. A distributed arrangement performs well only under one coordinator’s supervision. Manifestation does not establish independent durability or transfer.
18. A credential remains current after capabilities decay. Status and record do not establish present achievement.

13. Three Views and a non-container review envelope

Every material Formation claim should be reviewable through the universal minimum of three Views:

- the **Human View** examines experience, meaning, dignity, power, participation, accessibility, burden, and consequences for people;
- the **Scientific View** examines constructs, mechanisms, Evidence, uncertainty, alternatives, measurement, durability, transfer, and explanatory level; and

- the **Engineering View** examines realization, interfaces, dependencies, controls, records, failure modes, change, and operation.

The Views are not containers, professions, authorities, or competing truths. Any Projection across them must declare material loss. A Human description of coercion cannot be erased because an engineering metric improved. A scientific attribution cannot allocate legal Authority. A precise schema cannot make a weak formative claim true.

For a consequential attribution, a non-container review envelope may ask:

Accessible table projection: Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **Dimension:** Predicate
  - **Review question:** Is the claim about Formation, Learning, Instruction, Training, trained achievement, Education, Socialization, or another concept?
- **Dimension:** Subject
  - **Review question:** Which human, artificial, collective, institutional, distributed, or hybrid subject is claimed to bear the effect?
- **Dimension:** Boundary
  - **Review question:** What establishes the subject or Learner boundary and its continuity through change?
- **Dimension:** Time
  - **Review question:** Is the object an episode, activity, effect, continuing pattern, achievement, status, decay, or recovery?
- **Dimension:** Purpose
  - **Review question:** What developmental or performance condition was declared, by whom, and under what Authority?
- **Dimension:** Conditions
  - **Review question:** Which environment, exposure, practice, intervention, relationships, incentives, and constraints were material?
- **Dimension:** Mechanism
  - **Review question:** How is the effect linked to the conditions, and which alternatives remain plausible?
- **Dimension:** Manifestation
  - **Review question:** What changed at the claimed level under relevant variation?
- **Dimension:** Durability and transfer
  - **Review question:** Does the change persist, depend on support, and extend beyond the observed cases?
- **Dimension:** Power
  - **Review question:** Who could compel, refuse, contest, evaluate, revise, or exit, and what vulnerabilities mattered?
- **Dimension:** Affected horizon
  - **Review question:** Who participated, contributed, benefited, or bore burdens without participating?
- **Dimension:** Evidence
  - **Review question:** What Grounds and Evidence support each claim, with what uncertainty and dissent?
- **Dimension:** Views and loss
  - **Review question:** What do the Human, Scientific, and Engineering Views preserve or omit?
- **Dimension:** Status and lineage

- **Review question:** Who assessed or authorized what, which procedure governs change, and when must the claim be reviewed?

This envelope is not a certification, package anatomy, ontology, implementation form, or universal workflow.

14. Development requirements

Implementation-facing work should preserve the following distinctions without silently making this inventory canonical schema:

14.1 Represent claims, not just interventions

Where consequence warrants it, records should be capable of distinguishing subject, predicate, purpose, condition, activity, mechanism, effect, achievement, assessment, status, Evidence, uncertainty, authority, affected horizon, and revision lineage.

14.2 Separate declaration, practice, manifestation, and status

A stated value, required procedure, completed exercise, observed performance, durable disposition, educational attribution, and recognized credential are different objects.

14.3 Inspect formative conditions

Metrics, interface prominence, defaults, permissions, response timing, role structures, incentives, examples, sanctions, and tolerated violations should be reviewed for plausible formative effects without assuming that an effect occurred.

14.4 Support plural formative routes

Where Training is appropriate, systems may support practice, variation, feedback, spacing, reflection, exemplars, correction, and transfer tests. Where other routes are appropriate, systems must not force them into Training terminology.

14.5 Preserve developmental history and alternatives

Current conduct alone can conceal memorization, coercion, retrieval, supervision, replacement, environmental redesign, model change, member Learning, institutional change, dependency, or decay.

14.6 Detect failure, drift, and deskilling

Evaluation should distinguish Knowledge without enactment, enactment dependent on supervision, rule-following without transfer, compliance under coercion, local optimization that harms a wider system, habits persisting after their justification fails, capability decay, and deskilling through substitution.

14.7 Distinguish correction, repair, remedy, and Learning

Feedback does not establish reception; correction does not establish repair; repair does not establish remedy; and none establishes Learning. Records should preserve which response occurred and at which level.

14.8 Govern formative power

Systems should make visible who selected the objective, what Authority applies, which conception of desirable conduct is encoded, who may challenge it, which alternatives exist, and who bears consequences. Actual rights, consent, duties, remedies, and decisions remain with competent Domain or external authority.

14.9 Evaluate the lowest adequate subject and the cooperative unit

Individual assessments are insufficient when failures arise from coupling, incentives, interfaces, institutions, or distributed control. Whole-system claims are excessive when a component-level explanation is adequate.

14.10 Preserve governed challenge

Participants and affected subjects need inspectable challenge routes proportionate to consequence. Evidence challenges claims and decisions; artifact change, suspension, authorization, or remedy follows competent procedure.

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15. Accepted answers and live questions

The v0.1 paper presented eight questions as uniformly open. The accepted baseline resolves part of each while preserving narrower inquiry.

Accessible table projection: Each source row follows as a labelled record; all source headers and cell wording are preserved.

- **v0.1 question:** Can every Intelligence be educated?
  - **Accepted minimum:** Intelligence does not entail Education. Education requires its own subject-level Evidence; substrate alone does not decide.
  - **Live remainder / owner:** Minimum literal attribution conditions across artificial, collective, distributed, and hybrid cases remain empirical, conceptual, Domain, and later standards work.
- **v0.1 question:** Is reflective capacity necessary?
  - **Accepted minimum:** Education includes capacity to examine and revise the frame; that capacity may be distributed rather than internal to each component.
  - **Live remainder / owner:** How distributed reflection is realized and evidenced remains open by Domain and subject topology.
- **v0.1 question:** What distinguishes Education from successful Socialization?
  - **Accepted minimum:** Education is broad, integrated, participatory, and challengeable; Socialization may reproduce harmful orders.
  - **Live remainder / owner:** Operational assessment remains Domain- and authority-sensitive.
- **v0.1 question:** Which habits are universal to SymK?
  - **Accepted minimum:** No universal virtue catalogue is accepted. Thin Cooperation and semantic non-substitutions control.
  - **Live remainder / owner:** Domains may justify named dispositions; later constitutional work may consider narrower duties.
- **v0.1 question:** When is Training coercive?
  - **Accepted minimum:** Coercion, Training, consent, Authority, legitimacy, effect, and record are distinct.
  - **Live remainder / owner:** Legal, professional, ethical, institutional, and Domain authorities own thresholds, permissions, protections, and remedies.
- **v0.1 question:** Can an institution be ill-mannered?
  - **Accepted minimum:** Institutions can bear relational, power, Formation, and possibly Learner predicates when supported at that level.
  - **Live remainder / owner:** “Ill-mannered” remains contextual shorthand, not a universal technical status.
- **v0.1 question:** How should educational decay be represented?
  - **Accepted minimum:** Decay, drift, deskilling, unlearning, current status, and record are distinct and must remain visible.
  - **Live remainder / owner:** Formal models, review periods, measures, and thresholds belong downstream.
- **v0.1 question:** Who educates the educator?
  - **Accepted minimum:** No formative position is semantically immune from Evidence, challenge, or competent review.

- **Live remainder / owner:** Actual governance bodies, standing, appeals, duties, remedies, and jurisdiction remain later or external decisions.

These remainders do not create new deferred questions. They preserve accepted DQ-004, DQ-013, DQ-019, and related downstream ownership.

16. Value, consequence, and revision

Education, Training, Learning, and Formation enter a branching, defeasible, and recursive value relation. A formative purpose may answer one Question while creating another. The same achievement can benefit one subject and burden another. A trained capability can enable responsible action or amplify harm. An educational program can support participation while assimilating difference. Observations and claims about consequences can challenge premises, motivate frame review, support status change through competent procedure, indicate repair, or support ending an undertaking; none automatically revises or invalidates another object.

Value claims therefore require an evaluator or evaluating process, object, criterion, Scope, Context, affected horizon, time, Evidence, uncertainty, and Authority. “Improved,” “responsible,” “successful,” and “educated” are not interchangeable. No learning metric, credential, benchmark, instructor judgment, participant satisfaction, or system outcome alone settles Value.

When Evidence challenges a formative claim, the claim’s premises, Applicability, alternatives, consequences, assessment, and decisions should be reviewed. The governed artifact or Authority source changes only through competent procedure. Urgent adverse Evidence triggers urgent consideration by competent operational or external Authority; no emergency duty, power, remedy, or jurisdiction is created by this explanatory paper.

17. Conclusion

SymK cannot assume that Intelligence plus Knowledge yields Cooperation. It also cannot fill that gap by making Education constitutional to every cooperative or persistent relation, or by treating Training as Education’s universal mechanism.

The foundational concept is Formation: neutral, temporally extended shaping or reorganization at an identified subject level. Learning, Training, and Education are distinct supporting concepts within that system. Training is directed structured formative activity; Learning is comparatively durable experience-linked change; Education is broad, integrated, participatory, and challengeable Formation. They can overlap. None universally entails either of the others.

This structure preserves the v0.1 paper’s deepest concern: systems shape participants through more than declared rules. It makes that concern more precise. Conditions are not effects. Activities are not achievements. Records are not current reality. Member change is not system Learning. Influence is not Education. Persistence is not responsible development. Cooperation is not final Value.

> **A material formative claim must identify what changed, at which subject level, through which conditions and mechanism, with what durability, transfer, Evidence, alternatives, power relations, affected consequences, and competent status.**

That obligation is explanatory and review-oriented. It does not select an axiom, curriculum, authority, package, schema, metric, threshold, policy, or implementation.

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The references preserve the v0.1 inquiry lineage. They support historical, philosophical, or empirical context; they do not independently determine SymK semantics, constitutional force, authority, universal pedagogy, or an actual Formation attribution.

Document governance note

This review draft has completed Stream F cross-paper integration analytically through `SYMK-2X-EV-166` and should evolve only through explicit SymK governance. Revisions should distinguish literature corrections; changes to accepted semantic explanation; changes to paper-level examples or review questions; proposed constitutional changes; and downstream implementation choices.

The concepts, counterexamples, object distinctions, explanatory obligations, and review envelope in this paper are Human-facing evidence and constraints. No implementation should silently promote them into canonical package anatomy, identifiers, ontology, schema, metric, workflow, policy, curriculum, authority allocation, right, duty, standing, remedy, or operational decision.